

Structurizr Diagrams

This folder contains Structurizr DSL source for structured architecture and workflow diagrams.

Source

- `workspace.dsl` is the source of truth.
- Exported images should be written to `../exports`.
- Markdown documents should embed exported SVG or PNG files after rendering.

View Keys

View key	Replaces or supports
<code>SystemContext</code>	Solution Architecture - System Context Mermaid diagram.
<code>ContainerArchitecture</code>	Solution Architecture - major containers and integration paths.
<code>ConceptualDataModel</code>	Data Architecture - Conceptual Data Model Mermaid ER diagram.
<code>NominationCleanApprovalFlow</code>	Solution Architecture - Core Workflow and Business Architecture - Nomination Lifecycle.
<code>NominationHRBPReviewFlow</code>	HRBP branch of the nomination lifecycle.
<code>ModelLifecycle</code>	AI/ML Architecture - Model Lifecycle Mermaid diagram.
<code>PayrollIntegrationFlow</code>	Integration Architecture - Payroll Integration Mermaid diagram.
<code>AzureDeployment</code>	Solution Architecture - Deployment Topology Mermaid diagram.

Open workspace.dsl in the Structurizr server (Tomcat) — proven steps

Structurizr **Lite** (`structurizr/lite`) is discontinued — that image now only prints a migration notice and exits. The replacement is the consolidated `structurizr/structurizr` image run with the `local` argument, which serves the workspace from an embedded **Tomcat** server on port 8080.

1. Start the server

Run this in **PowerShell** (single line; adjust the path only if the repo moves). It mounts the folder that contains `workspace.dsl` and starts Tomcat detached:

```
docker rm -f structurizr-local 2>$null
docker run -d --name structurizr-local --user 0:0 --dns 127.0.0.1 -e STRUCTURIZR_NETWORK_TIMEOUT=2000 -p
```

2. Wait for Tomcat to be ready

```
docker logs -f structurizr-local
```

Wait for the line `Tomcat started on port(s): 8080` (also `Started ... in N seconds`). Then press `Ctrl+C` — that only stops the log tail; the container keeps running because it is detached. Startup takes ~10–40s.

3. Open it

```
http://localhost:8080
```

It redirects to <http://localhost:8080/workspace/1> (the diagram index). Individual diagrams are at <http://localhost:8080/workspace/1/diagrams#<ViewKey>> (e.g. <http://localhost:8080/workspace/1/diagrams#SystemContext>).

4. Export a diagram to SVG/PNG

Open a diagram, then in the top toolbar click the **Export** icon (far-right, picture/landscape glyph) → **Export as SVG** (or PNG). Save into `../exports` using the view key as the filename (see the table above). Each SVG export also produces a small `*-key.svg` legend file.

5. Stop the server

```
docker rm -f structurizr-local
```

Why the extra flags (do not remove them)

These were all required to get it running on this machine; stripping any one reintroduces a failure we already diagnosed:

Flag	Reason
<code>structurizr/structurizr:<tag> local</code>	Lite is discontinued; <code>local</code> is its replacement mode. Use a <code>non--playwright</code> tag — the <code>-playwright</code> variant hangs on headless-Chromium init at startup.
<code>--user 0:0</code>	The container writes its log + Lucene search index into the mounted folder. Without root it fails with <code>Permission denied / write.lock AccessDenied</code> and never starts. Use numeric <code>0:0</code> (the image is distroless, so <code>--user root</code> by name fails: "no matching entries in passwd file").
<code>--dns 127.0.0.1</code> and <code>-e STRUCTURIZR_NETWORK_TIMEOUT=2000</code>	On startup the server makes an outbound "theme/version" network call. Behind the Cisco AnyConnect VPN that call black-holes and the boot freezes at the <code>Themes:</code> log line for minutes. Breaking DNS + shortening the timeout makes it fail fast and continue offline.
<code>-p 8080:8080</code>	Publishes Tomcat's port to the host.
<code>-v "C:/...:/usr/local/structurizr"</code>	Mounts the folder that contains <code>workspace.dsl</code> . Use forward slashes to avoid the Windows drive-letter colon clashing with the mount separator. Point at the folder, not the file.

Troubleshooting

- `ERR_CONNECTION_REFUSED` → nothing is listening; the container exited or isn't started. Check `docker ps` and `docker logs structurizr-local`.
- `ERR_EMPTY_RESPONSE` / **page won't load** → Tomcat hasn't finished starting; wait for the `Tomcat started on port(s): 8080` log line.
- **Log frozen at Themes:** → the startup network call is hanging; ensure the `--dns 127.0.0.1` and `STRUCTURIZR_NETWORK_TIMEOUT` flags are present (VPN issue).
- **A hidden `.structurizr/` folder appears here** (logs + search index) — that is expected and safe to gitignore.

Validate or Export with Structurizr CLI

If the Structurizr CLI is installed or the CLI JAR is available:

```
java -jar structurizr-cli.jar validate -workspace Documentation_Codex\diagrams\structurizr\workspace.dsl
```

Example export command:

```
java -jar structurizr-cli.jar export -workspace Documentation_Codex\diagrams\structurizr\workspace.dsl
```

The CLI can export intermediate formats such as PlantUML, Mermaid, DOT, or JSON depending on the installed version. For polished SVG/PNG output, Structurizr Lite is usually the easiest first step.

Markdown Embedding Pattern

After exporting a diagram to SVG:

```

```

For documents inside volume folders, adjust the relative path:

```

```

The exported image is what Markdown displays. The `.dsl` file remains the maintainable source.